Cross-Modal Fusion and Classification of Hyperspectral and Panchromatic Remote Sensing Images With Deep Learning and Multiscale CRFs

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***Abstract*—This article presents a multiresolution supervised method for the joint classification and fusion of panchromatic and hyperspectral data. This problem gains relevance especially in light of recent and upcoming missions—such as PRISMA, EnMAP, and CHIME—that include both hyperspectral and panchromatic sensors. The method combines deep learning (DL) with a hierarchical conditional random field (CRF) to extract and fuse multiresolution features. The objective is to generate a land cover map with the fine spatial resolution of the panchromatic data, while exploiting the rich spectral information from the hyperspectral channels. The proposed DL framework includes a two-branch encoder that separately processes spatial and spectral data from the two input sources, followed by a decoder that performs a first stage of multiresolution fusion. The considered architectures range from the traditional fully convolutional encoder–decoder model to cross-modal attention mechanisms using vision transformers (ViTs). The network posterior probability estimates, derived at both the panchromatic and hyperspectral levels, are subsequently integrated in a multiresolution CRF that incorporates a clustering step, enabling the integration of both local and long-range spatiospectral relationships in both the hyperspectral and panchromatic domains. The proposed method has been experimentally validated with PRISMA data for land cover mapping in the Emilia-Romagna and Lombardy Regions, Italy, and in the Castilla–La Mancha area, Spain.**

***Index Terms*—Conditional random field (CRF), cross-modal multiresolution fusion, fully convolutional network (FCN), hyperspectral classification, semantic segmentation, vision transformers (ViTs).**

I. Introduction

THIS article addresses the complex problem of the supervised dense classification—or semantic segmentation—of multiresolution hyperspectral-panchromatic data, with the aim of generating a classification map at the resolution of the panchromatic channel while exploiting the richness of the spectral information provided by the hyperspectral channels. This multiresolution problem is very promising from the point of view of thematic information extraction, precisely thanks to the opportunity to benefit from both spatial and spectral information captured by the two different sensors. The practical relevance of this setting is reinforced by recent and upcoming Earth observation missions—such as PRISMA, EnMAP, and ESA's CHIME—which include co-registered hyperspectral and panchromatic instruments [1], [2].

Unlike single-resolution hyperspectral data [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], the classification of coregistered panchromatic and hyperspectral data across multiple resolutions is a new problem that has not been extensively explored in the literature.

The proposed multiresolution fusion technique for classification is based on methodological contributions stemming from deep learning (DL) and probabilistic graphical models (structured output learning). Great effectiveness in satellite image classification has been demonstrated by deep neural networks, typically based on the interconnection of multiple vision transformer (ViT) or CNN subnetworks. The latter are characterized by an intrinsically multiresolution structure, thanks to the presence of pooling layers that perform spatial subsampling processes, and convolutional layers, whose spatial windows determine the extraction of levels of representation at distinct spatial scales (representation learning) [15]. This structure, typical of 2-D CNNs on pixel networks, then naturally extends to datasets associated with topologies of different sizes, as well as the 3-D datacube associated with a hyperspectral image. Such extensions make use of 3-D-CNN or 1-D convolutions along the channel set (equivalent to the wavelength axis; 1-D-CNN) [16]. Furthermore, recent DL approaches have demonstrated state-of-the-art performance in cross-modal remote sensing image analysis, leveraging their ability to learn robust, modality-invariant representations across heterogeneous data sources such as optical (hyperspectral, multispectral), SAR, and LiDAR [17], [18], [19], [20], [21], [22], [23].

In this context, the proposed method is based, first, on CNN-type architectures (including also fully convolutional networks (FCNs) [24]) and ViTs [25], dedicated to the estimation, on the panchromatic network, of the posterior

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probabilities of thematic classes, given both hyperspectral and panchromatic observations. To this end, the neural architecture aims to integrate the hyperspectral and panchromatic data at the corresponding spatial resolution levels and generate estimates of the posterior probabilities.

To effectively train such deep networks, a very large number of precisely annotated training samples is usually needed. This condition is easily verified in the application to benchmark data made available for international scientific contests (see, for example, [26]), but it often turns out to be a critical restriction in real-world applications, in which the available training sets are usually composed of a relatively limited number of samples or spatially disjoint regions assigned to the classes [27]. More detailed information on the spatial structures in the scene (objects and regions) is therefore often absent from the training set. This negatively influences the performance of a neural learning model trained on such data and constitutes an operational limitation. The technique proposed here also aims to address, in the context of multiresolution panchromatic-hyperspectral classification, this criticality related to the requirements in terms of quantity and quality of the training set. To this end, the proposed method integrates the approach based on deep neural networks with a probabilistic graphical model, namely a conditional random field (CRF). The technique extends the previous method in [27], which focused on the classification of single-resolution RGB optical aerial images with a spatial resolution of a few centimeters, to the case of the classification of multiresolution panchromatic-hyperspectral satellite images. The key idea consists in mitigating the impact of the training set insufficiency by incorporating, with an innovative CRF model, information associated with not only local but also long-range spatial–spectral relations. First, the graph topology and the energy function of the CRF model in the proposed approach are formulated with the goal of characterizing the multiresolution information associated with the input data. Then, it is assumed that the neural network is trained with the available training data, albeit limited in quantity and/or quality. Ideally, an effective approach to compensate for the impact of such a training set on accuracy would be to use a fully connected CRF, i.e., a probabilistic graphical model that characterizes the interactions between all pairs of pixels in the considered grid—a model whose computational complexity would be intractable. A traditional Markov random field (MRF) or CRF model characterizes relationships on a local basis, i.e., in terms of a system of neighborhoods [28], [29].

The CRF model proposed here introduces a graph that includes not only the local relations defined by the neighborhood scheme but also relations between a set of additional virtual nodes aimed at representing long-distance dependencies in both the hyperspectral and the panchromatic images. Such virtual nodes are defined on the basis of a clustering result of the computed activations of the neural network in the set of all its hidden layers. Specifically, on the resulting composite graph, which extends to both pixel grids at the panchromatic and hyperspectral spatial resolutions and to the set of virtual nodes, a CRF is defined that characterizes the interactions: 1) between neighboring pixels; 2) between virtual nodes interconnected with a full mesh; 3) between pixels and virtual nodes interconnected with a dedicated neighborhood scheme; and 4) among pixels in the lattices across the two considered resolutions. In this way, long-range relationships are captured by the model, while retaining a computationally feasible formulation. The application of state-of-the-art graph-based minimum energy algorithms, involving graph cuts [30], ensures convergence to solutions characterized by strong optimality conditions. Methodologically, the proposed technique aims to capture and integrate into a probabilistic decision process the information extracted from the network at various spatial scales, taking advantage of both the great potential of deep neural networks in learning from high-dimensional datasets, information at different levels of abstraction, and the ability of probabilistic graphical models to optimize classification results, explicitly formalizing the dependencies associated with the input data.

The proposed framework assumes that the panchromatic and hyperspectral inputs are spatially aligned prior to classification and multiresolution fusion. While image registration is not a component of the method itself, reliable techniques are available for the correction or reduction of possible geometric misalignment [31], [32].

The main novel contributions of this work are twofold: 1) the definition of a semantic segmentation framework based on DL and stochastic models that jointly operates on hyperspectral and panchromatic imagery in order to generate a classification map at the higher spatial resolution of the panchromatic sensor; 2) the development of a novel CRF model on a multiresolution graph topology, integrated together with deep neural networks, to model the interactions both within and across the hyperspectral and panchromatic grids and to address the multiresolution fusion of hyperspectral and panchromatic data even in the presence of limited or unevenly distributed training samples. Regarding point 1), to the best of our knowledge, this work is the first involving the cross-modal fusion of panchromatic and hyperspectral images for the generation of a semantic segmentation output at the panchromatic scale. We also recall that a preliminary single-resolution version of this work was published in the conference paper [33]. As compared to that early work, the present article both differs significantly in methodology, which incorporates here DL- and CRF-based multiresolution modeling as well as a ViT formulation, and presents an expanded experimental validation, which involves an additional dataset and further experimental comparisons.

The article is organized in the following way. Section II provides an overview of the state of the art with respect to DL methods for hyperspectral image classification based on multimodal information; Section III presents the proposed methodology. The results of the experimental validation conducted with the proposed framework and the comparison with the results obtained by state-of-the-art techniques are described and discussed in Section IV. Finally, conclusions and perspectives of the proposed technique are reported in Section V.

II. Related Work

In this section, previous approaches to hyperspectral image semantic segmentation are reviewed, with a focus on methods involving multimodal information fusion.

Hyperspectral image classification is a highly active and investigated area of remote sensing, for which effective techniques provide very accurate results. In particular, several approaches for supervised classification of hyperspectral data based on the analysis of spatial and spectral information have been proposed [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14] based on different image processing techniques. The methods in [4] and [5] develop architectures combining MRFs and dictionary learning or support vector machines (SVMs), respectively, while [6] combines SVMs and DL techniques. The architectures presented in [7], [8], [9], [10], [11], [12], and [13] are also based on DL models, specifically [9], [10] involving the combination of neural networks and CRFs, [11], [12] graph convolutional network models, [13] graph transformer networks, and [34], [35], [36], [37], [38] convolutional neural networks integrating the analysis of spectral information to the more traditional analysis of spatial information guaranteed by 2-D convolutional layers.

Other approaches for supervised classification of hyperspectral images include preliminary classification methods, such as superpixels [39], [40], [41], [42], in some cases to overcome the scarcity of available input information [39] typical of applications related to land cover mapping from remote sensing.

At the same time, unlike the above-mentioned classification of single-resolution hyperspectral data, joint panchromatic-hyperspectral multiresolution classification is a very new problem, substantially not addressed in the literature so far.

Panchromatic-multispectral classification techniques have been developed in [43] and [44] by means of MRF-based approaches, but with reference only to sensors characterized by a limited number of spectral channels and without taking into account the potential and challenges of the hyperspectral data. Other panchromatic-multispectral classification methods present in the literature are based on feature fusion, DL models [45], filtering-based intensity modulation [46], and pyramid representations [47]. The latter have also been used in the conjunctive RGB-hyperspectral classification [48].

Several recent methods have been proposed for joint hyperspectral-LiDAR classification. These approaches combine the spectral richness of hyperspectral imaging with the spatial 3-D precision of LiDAR to enhance land cover analysis by leveraging both spectral and structural-geometric data. Various data fusion techniques based on multimodal feature extraction by ensemble learning [49], and by DL have been developed, such as cross-modal attention fusion networks on multiscale pyramidal structures [18], superpixel-based multilayer perceptrons (MLPs) [50], spatial–spectral saliency reinforcement networks [51]. These methods usually involve input information at the same scale and with the same image size.

A related task of multiresolution fusion of hyperspectral and panchromatic data is hyperspectral pansharpening (PS). However, hyperspectral PS methods [52], [53] address the problem of multiresolution fusion of hyperspectral and panchromatic channels with a different objective, oriented to generate not a classification result but a simulation of the hyperspectral data on the pixel grid of the panchromatic channel. These techniques do not consider the relationship between the hyperspectral and panchromatic measurements and the spatial field of thematic class labels (e.g., land cover or use), do not use training samples for the classes, and determine, on the basis of concepts of signal processing and without a goal of supervised classification, a transformed image that is not optimized in order to correctly discriminate the classes.

III. Methodology

A. Overview of the Proposed Methodology

The proposed methodology performs multiresolution and multispectral image data fusion for semantic segmentation by integrating a spatial–spectral multiresolution neural network architecture, clustering, and a multiresolution CRF model (see Fig. 1). This approach, in the following referred to as HPMC (hyperspectral-panchromatic multiresolution classifier), leverages the high spatial resolution of panchromatic data and the rich spectral detail of hyperspectral data, with the goal of combining their complementary strengths.

The problem involves image data of the same scene acquired at multiple resolutions and with different spectral information. In particular, the input data are panchromatic, $\mathcal{X}^p \in \mathbb{R}^{1\times H\times W}$ and hyperspectral, $\mathcal{X}^h \in \mathbb{R}^{C_h\times H/\rho\times W/\rho}$ acquisitions, where $H \times W$ is the size of the pixel lattice at the resolution of the panchromatic image, ρ is the ratio between the spatial resolutions of the two considered pixel lattices, and C_h is the number of hyperspectral channels. H, W, and ρ are supposed to be such that the height H/ρ and width W/ρ of the pixel grid of the hyperspectral image are integers.

The multiresolution fusion is first addressed by the proposed neural network, which employs a two-branch encoder. One branch processes the high spatial resolution panchromatic data, while the other processes high spectral resolution hyperspectral data. Features from both branches are fused in a bottleneck layer, which aligns their resolutions for subsequent decoding.

Two different neural architectures have been developed in the framework of the proposed approach, one based on FCNs and one based on ViTs. The idea is to leverage the cross-modality feature fusion capabilities of the two different models, either through depthwise pointwise convolutions and spectral max pooling, compressing spectral information into spatially meaningful features, or through attention mechanisms, processing and aligning information across modalities by assigning different weights to different modality interactions.

Afterward, the feature maps containing the panchromatic and hyperspectral data processed by the network are extracted, at their different native resolutions—thus at different blocks of the decoder and the two-branch encoder. Clustering is applied separately to the feature maps extracted on the two lattices using k-means, with the goal of identifying spatial–spectral similarities across the imaged scene. This step favors connections between similar pixels across the two multiresolution images, independently of their spatial proximity.

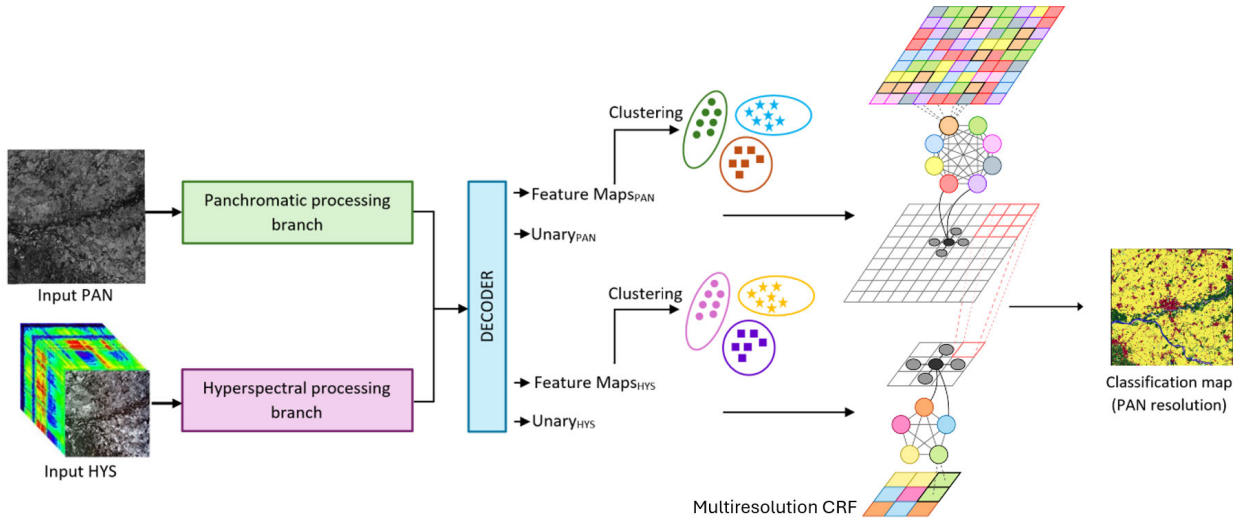


Fig. 1. Overall architecture of the proposed methodology.

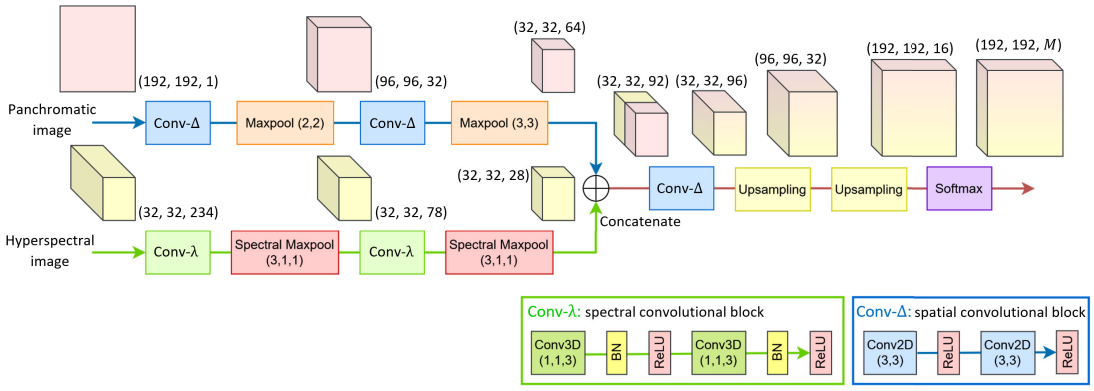


Fig. 2. λFCN: proposed DL architecture based on FCNs for hyperspectral-panchromatic image fusion.

To further refine the segmentation and fuse the multimodal information, a case-specific multiresolution CRF is defined. The CRF energy function incorporates unary potentials derived from network predictions and pairwise potentials to enforce spatial smoothness and label consistency. These pairwise potentials connect pixels locally, clusters globally, pixels to their nearest clusters, and pixels between the two multiresolution data, capturing hierarchical relationships.

Overall, the methodology integrates multiresolution data fusion with DL-based feature extraction, clustering, and CRF-based modeling of long-range spatial dependencies. Further details are provided in Sections III-B and III-C.

B. Spatial–Spectral Neural Architecture

Two encoder–decoder architectures are proposed for the first step of the multiresolution fusion: 1) a FCN-based architecture and 2) a ViT-based architecture. Their descriptions are provided in Sections III-B1 and III-B2.

1) FCN-Based Architecture: The first proposed neural network architecture belongs to the family of FCNs [24]. In this case, the architecture is designed for a semantic segmentation task that integrates two different multiresolution data sources: panchromatic and hyperspectral channels. The architecture is based on a U-Net [54] and extended for panchromatic-hyperspectral image processing. It integrates both spectral and spatial feature extraction through a combination of standard 2-D convolutions, specialized spectral convolutions, and multiscale upsampling and downsampling layers. The network consists of an encoder–decoder structure with skip connections, allowing it to capture both low-level and high-level features while preserving spatial information. The overall structure is presented in Fig. 2.

The encoder is divided into two separate branches, which process panchromatic and hyperspectral images separately. The first branch processes the high-resolution panchromatic data $\mathcal{X}^p$. The input passes through a series of spatial convolutions with batch normalization and rectified linear unit (ReLU) activations, and downsampling max pooling layers. The encoder progressively downsamples the panchromatic image while increasing feature depth.

Meanwhile, the second branch processes the coarser spatial resolution hyperspectral data $\mathcal{X}^h$, applying a spectral feature extraction block consisting of depthwise pointwise convolutions followed by max pooling along the spectral dimension,

to compress spectral information before merging it with the spatial features. We emphasize the role of depthwise convolutions, whose goal is to extract a meaningful representation from the hyperspectral channels, to benefit from their rich spectral content. In particular, this processing stage compresses the spectral information while retaining spatial features, with the aim of condensing multichannel spectral data.

The features extracted from both channels are fused in a bottleneck layer, which learns the combined representations of spatial and spectral data. In order to have a match between the two original spatial resolutions, the sequence of pooling layers have an overall size which is a multiple of the original resolution ratio, $\nu \cdot \rho$ with $\nu \in \mathbb{N}$. The bottleneck consists of two convolutional blocks with batch normalizations and ReLU activations.

The decoder reconstructs the classification map by progressively upscaling the features. Upsampling is performed using either bilinear interpolation or transposed convolutions, followed by feature concatenation with the corresponding encoder layer and processing with convolutional layers, which refine the features while retaining spatial details. The final classification map is generated at the spatial resolution of the panchromatic image. The network includes dropout layers for regularization, in order to prevent overfitting.

By combining spectral feature extraction with a U-Net-inspired two-branch spatial processing pipeline, the proposed hybrid spectral–spatial processing network is aimed at capturing different levels of abstraction from both spectral and spatial characteristics, with the goal of panchromatic-hyperspectral image fusion and classification. The FCN-based architecture within the proposed method will be referred to as λFCN in the following (where λ emphasizes the focus on spectral modeling).

2) ViTs-Based Architecture: The second proposed neural backbone is based on ViTs. It is a lightweight extension to hyperspectral-panchromatic image fusion and classification of the network proposed in [17] for RGB-DSM (digital surface model) data. As in the previous FCN-based architecture, first, two branches based on CNNs separately extract multiscale hyperspectral and panchromatic features, which are then fused by a crossmodal multiscale transformer. These extracted features are further enhanced by a cross-scale attention-based transformer to refine the multiscale fusion of multimodal features and filter the representations propagated through our multiscale skip fusion strategy. The output is fed into the decoder for upsampling and classification. This cross-modal attention network will be referred to in the following as λCMAtt (where "CMAtt" stands for cross-modal attention). Its architecture is shown in Fig. 3.

The encoder branch that processes the panchromatic data follows a hierarchical structure with three convolutional blocks, each containing two convolutional layers followed by batch normalization and ReLU activation. The extracted features undergo progressive downsampling using maxpooling operations, which capture spatial hierarchies while retaining key structural details. Similarly, the encoder branch that processes the hyperspectral imagery employs a parallel architecture but processes a 3-D input with 234 spectral channels.

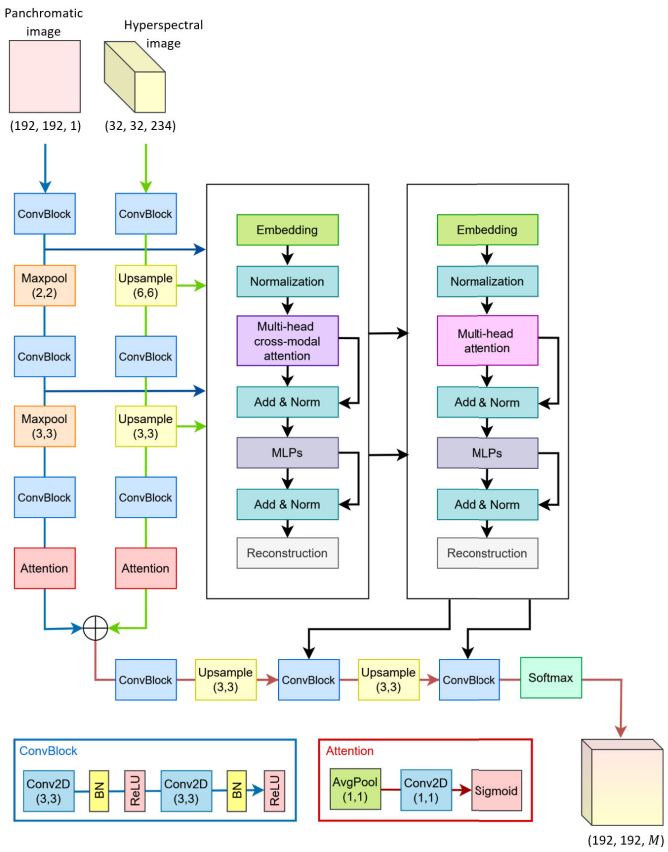


Fig. 3. λCMAtt: proposed DL architecture based on ViTs for hyperspectral-panchromatic image fusion.

Unlike the panchromatic encoder, it incorporates 3-D max pooling to reduce spectral dimensionality while preserving the spatial dimensions. Hence, at the output of the encoder, the processed patches deriving from panchromatic and hyperspectral data have the same sizes.

The fusion module employs a channel-wise attention mechanism leveraging global average pooling, 1×1 convolutions, and sigmoid activations to refine the extracted features by emphasizing the most informative channels. In parallel, the network uses transformer-based skip connections. These transformers facilitate feature aggregation between the results of the two branches of the encoder by learning long-range dependencies. The output from these transformers is then further refined using attention-weighted element-wise multiplication, which favors that the most relevant spectral–spatial features are retained before decoding.

The decoder structure mirrors the encoder, gradually reconstructing the fused image using bilinear upsampling and skip connections. It first recovers mid-level features by upsampling and integrating with earlier-stage panchromatic and hyperspectral encoder outputs, incorporating outputs from the transformer modules for additional refinement. This is followed by convolutional and batch normalization layers to refine feature representation before the final output.

Overall, λCMAtt combines CNN-based hierarchical feature extraction, attention-enhanced fusion, and transformer-based cross-domain learning. By leveraging these components, it

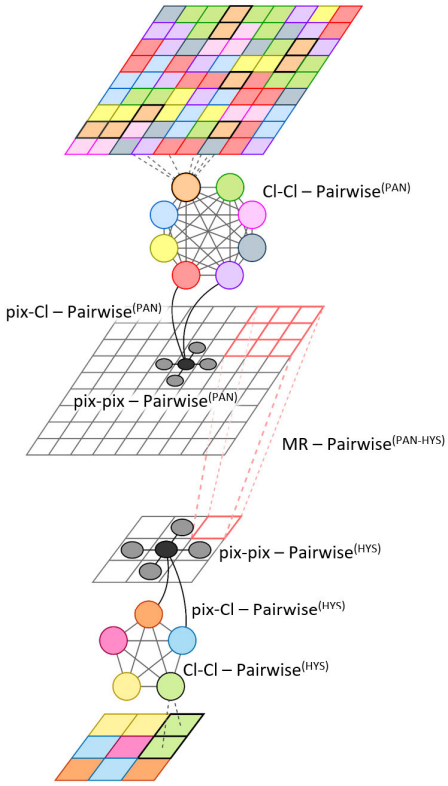


Fig. 4. Cluster-based multiresolution CRF in the proposed method connects the pixels at the two spatial resolutions, as well as each pixel (black circle) to its neighbors (gray circles) in each single resolution lattice, and with the h most similar clusters (colored circles). In turn, the clusters are fully connected.

favors high-resolution spatial detail retention while preserving spectral information for panchromatic-hyperspectral image classification purposes.

C. Cluster-Based Multiresolution CRF

The aforementioned neural architecture leverages the multiresolution information at its native resolution, producing feature maps at multiple resolutions which depend on the original fine resolution panchromatic channel of the image, on the coarse resolution hyperspectral channels, or on their combination (in the decoder).

The multiresolution CRF developed in the proposed approach leverages the feature maps ($\mathcal{F}^p$ and $\mathcal{F}^h$) and posterior probabilities produced at the native resolutions of the panchromatic and hyperspectral channels by either one of the proposed neural architectures. The set derived from hyperspectral data processing is the output of the hyperspectral branch of the encoder, whereas the set at the fine resolution of the panchromatic image is the output of the decoder of the network.

The developed CRF model (shown in Fig. 4) is a multiresolution extension, aimed at the fusion and classification of panchromatic and hyperspectral images, of the methodology proposed in [27] for the case of single-resolution aerial optical imagery. It incorporates a clustering partition that defines a set of virtual nodes in order to capture long-range dependencies. Specifically, k-means clustering is applied separately to the multiresolution feature maps $\mathcal{F}^p$ and $\mathcal{F}^h$ to identify k^p and k^h clusters, respectively. This step is aimed at joining similar pixels all over the feature maps in the same cluster, thus allowing connections through points at any distance on the image itself. The result is two sets of clusters $\mathcal{C}^p$ and $\mathcal{C}^h$, each associated with either the panchromatic or hyperspectral resolution.

The random fields of observations and labels are denoted $\mathcal{X}^\star = \{f_s^\star, x_c^\star\}_{s\in\mathcal{S}^\star, c\in\mathcal{C}^\star}$ and $\mathcal{Y}^\star = \{y_s^\star, y_c^\star\}_{s\in\mathcal{S}^\star, c\in\mathcal{C}^\star}$, respectively. The $\star$ represents the two possible different input lattices at panchromatic or hyperspectral resolution, respectively ($\star = p$ or h), $\mathcal{S}^\star$ is the corresponding image pixel lattice, and $f_s^\star$ is the value of the related feature map $\mathcal{F}^\star$ extracted by the network on pixel s. The feature vector $x_c^\star$ of cluster c is defined as the centroid of the feature vectors of the pixels belonging to that cluster ($c \in \mathcal{C}^\star$). It is important to note that, while the pixelwise posterior probability may be different for the two image lattices, the set Ω of thematic classes is the same at both resolutions.

The proposed multiresolution CRF model is aimed both at fusing short- and long-range spatial information within each pixel lattice and at fusing multiresolution information across the two lattices. Accordingly, its energy function is defined as follows:

$$\mathcal{E}(\mathcal{Y}|\mathcal{X}) = \mathcal{U}(\mathcal{Y}^p|\mathcal{X}^p) + \mathcal{U}\left(\mathcal{Y}^h|\mathcal{X}^h\right) + \zeta \sum_{\substack{s'\in\mathcal{S}^h \\ s\in\mathcal{N}(s')\subset\mathcal{S}^p}} V_{ss'}\left(y_s^p, y_{s'}^h\right). \tag{1}$$

Here, first, the $\mathcal{U}(\cdot)$ functions are single-resolution energy contributions on the individual panchromatic and hyperspectral lattices. Each of them is made up of pairwise nonzero clique potentials and integrates the aforementioned clustering partition

$$\begin{aligned}\mathcal{U}(\mathcal{Y}|\mathcal{X}) = &\sum_{s\in\mathcal{S}} D_s(y_s|f_s) + \lambda_{\mathcal{SR}} \sum_{\substack{r\in\partial s \\ s\in\mathcal{S}}} V_{sr}(y_s, y_r|f_s, f_r) \\ &+ \gamma \sum_{c\in\mathcal{C}} D_c(y_c|x_c) + \lambda_{\mathcal{CD}} \sum_{\substack{c,d\in\mathcal{C} \\ c\neq d}} V_{cd}(y_c, y_d|x_c, x_d) \\ &+ \lambda_{\mathcal{SC}} \sum_{\substack{s\in\mathcal{S} \\ c\in\partial s}} V_{sc}(y_s, y_c|f_s, x_c)\end{aligned} \tag{2}$$

where the $\star$ has been omitted to simplify the notation. D_s and D_c are the unary potentials for the pixel and cluster layers, and are computed as the pixelwise log-posterior probability predicted by each network. In the case of D_c, the posterior probabilities computed by the neural network on the individual pixels are averaged over cluster c. V_{sr} is a pixel pairwise potential that enforces spatial smoothness between neighboring pixels s and r ($s \in \mathcal{S}, r \in \partial s$, with $\partial s \subset \mathcal{S}$ being the neighborhood of pixel s). V_{cd} is a pairwise potential for each pair (c, d) of clusters. The cluster centroids are fully connected, and V_{cd} favors that similar clusters are assigned to the same class. V_{sc} is a pixel-cluster pairwise potential: while clusters are fully connected among themselves, each pixel s is connected to the h nearest cluster centroids—including the one it belongs to—based on a h-nearest neighbors (h-NN) criterion

[this set of h centroids is denoted $\bar{\partial}s$ in (2)]. This allows long-range information to propagate from the cluster layer to the pixel grid. All pairwise potentials $V(\cdot)$ in (2) are defined by a contrast-sensitive Potts function [55] to favor consistency in the labeling while simultaneously weighing on the similarity among the corresponding features.

The third term in (1) is a pairwise potential between the labels at the two spatial resolutions. $\mathcal{N}(s') \subset \mathcal{S}^p$ is the set of $\rho \times \rho$ pixels at panchromatic resolution corresponding to pixel $s' \in \mathcal{S}^h$ at hyperspectral resolution. The cross-resolution potential $V_{ss'}$ is defined as a Potts model [28] and enforces label consistency between each coarse-resolution hyperspectral pixel and the set of all fine-resolution panchromatic pixels falling in its spatial support. Here, the Potts model—and not its contrast-sensitive modification—is used because feature vectors associated with the hyperspectral and panchromatic lattices are not directly comparable (neither in spectral meaning nor in dimensionality). The λ-terms, γ, and ζ are positive weights that balance the various energy contributions.

According to this discussion, the proposed multiresolution CRF model in (1) is defined over a multiresolution graph that interconnects: 1) across the pixel lattices at the two spatial resolutions based on their common spatial support; 2) within each pixel lattice based on a Markovian neighborhood system; 3) between each pixel lattice and the corresponding clusters through a nearest neighbor criterion; and 4) among the clusters in a fully connected manner. From this perspective, instead of explicitly connecting all pixel pairs within each lattice, which would be prohibitively expensive for common hyperspectral and panchromatic scenes, the proposed model makes use of a cluster-based surrogate graph. Clusters play the role of "virtual nodes" summarizing groups of similar pixels; these virtual nodes are fully connected, and each pixel is connected only to a small number of nearest virtual nodes. Long-range pixel–pixel interactions are thus implicitly modeled by short paths (pixel → cluster → cluster → pixel) in this graph, yielding a sparse indirect approximation of dense pairwise connections with an affordable computational cost (see Section III-D). From this viewpoint, the component of the proposed CRF model that is associated with each lattice plays the role of an indirect approximation—through the clusters (virtual nodes)—of the effect of a fully connected CRF.

The energy function in (1) is minimized through the $\alpha - \beta$ swap graph cut method [56], which decomposes a multiclass inference problem into a sequence of binary ones. It converges to a local minimum with strong optimality properties [55], [57], [58].

D. Computational Complexity and Scalability

From a computational viewpoint, the training cost of the proposed framework, which includes both neural networks and a CRF model, is dominated by the neural components, whose complexity is comparable to standard encoder–decoder or transformer-based architectures with the same number of parameters (Tables I–III in Section IV). The CRF component is applied only at inference time to network activations and posteriors, and is implemented crop-wise ($n \times n$ pixels for the panchromatic image and $n/\rho \times n/\rho$ pixels for the hyperspectral image). This crop-wise strategy ensures that memory usage and runtime scale linearly with the scene size, enabling processing of large-scale images via tiling.

For each crop, the computational cost due to the CRF component is mainly due to: 1) k-means clustering of the multiresolution feature maps; 2) construction of the graph; and 3) α–β swap graph-cut inference. Since k-means is executed on a subset of pixels and with fixed numbers of clusters k^p and k^h, for the panchromatic and hyperspectral images, its cost scales linearly with the number of processed samples. The graph contains $n^2 + (n/\rho)^2 + k^p + k^h$ nodes. The edges include: 1) local pixel–pixel connections in each lattice; 2) fully connected cluster–cluster links; 3) pixel–cluster connections restricted to h nearest clusters per pixel; and 4) multiresolution pixel-pixel connections across the two lattices. As k^p, k^h, and h are expected to be much smaller than n^2 and $(n/\rho)^2$ (e.g., see Section IV), the resulting number of edges grows linearly with the number of pixels plus a negligible quadratic term in the number of clusters. In practice, the runtime overhead introduced by the multiresolution CRF in HPMC is not a bottleneck.

IV. Experimental Results

A. Dataset and Experimental Setup

The proposed HPMC technique was evaluated using three datasets collected from predominantly urban areas in Lombardy and Emilia-Romagna, Italy, and from a vegetated area in Castilla–La Mancha, Spain. The datasets comprise PRISMA images acquired in April 2021 for Lombardy and in July 2022 for Emilia-Romagna and Castilla–La Mancha. Each PRISMA acquisition includes a panchromatic image with a 5-m spatial resolution and a hyperspectral image containing 201 spectral bands at a 30-m spatial resolution.

The ground truth was derived based on regional land use archives: the 2021 DUSAF[1] for the Lombardy area, a 2020 regional archive for the urban areas of Emilia-Romagna,[2] and the 2020 SIOSE[3] archive for Castilla–La Mancha. The mapping classes aim to discriminate between different vegetated areas (cultivated fields, low and high vegetation), water bodies, and built-up areas (buildings and roads).

The datasets were divided into two separate subsets for the training and testing of the proposed multisensor fusion architecture. Specifically, in the case of either Lombardy or Castilla–La Mancha, the dataset consists of a PRISMA multiresolution image of a predominantly urban or vegetated area. This image was divided into two smaller and spatially disjoint subimages for training and testing, without any overlapping between the two sets. In contrast, the Emilia-Romagna dataset contains three images, which cover multiple urban centers and overlap in a small portion of the scene. In this case, two images were used for training, while the remaining one was reserved for testing after eliminating the overlapping zone.

[1] https://www.regione.lombardia.it/
[2] https://geoportale.regione.emilia-romagna.it/
[3] https://www.siose.es/

TABLE I

TEST-SET RESULTS FOR THE LOMBARDY DATASET. PER-CLASS VALUES ARE RECALLS. OVERALL ACCURACY (OA), RECALL, PRECISION, AND $F1$ SCORE ARE AVERAGED OVER THE CLASSES

Architecture	built-up	crop	tall veg.	grass	bare soil	water	OA	recall	precision	F1 score	# trainable param.
λFCN (proposed)	92.75	98.28	91.34	74.88	90.75	91.68	95.69	89.91	91.63	90.77	$17{\cdot}10^6$
LλFCN (proposed)	88.72	97.10	83.77	43.72	87.02	76.71	91.99	79.51	84.73	82.03	$23{\cdot}10^4$
λCMAtt (proposed)	81.88	92.81	74.67	47.98	76.06	91.69	87.50	77.52	70.28	73.72	$8{\cdot}10^6$
CFC-CRF [27] + λFCN	94.77	**99.00**	**93.01**	75.32	92.98	91.76	96.69	91.14	**94.63**	**92.85**	$17{\cdot}10^6$
CFC-CRF [27] + LλFCN	89.81	98.29	85.94	43.72	88.85	78.37	93.26	80.83	86.51	83.57	$23{\cdot}10^4$
CFC-CRF [27] + λCMAtt	82.39	93.81	76.09	43.53	75.74	90.82	88.24	77.06	72.06	74.48	$8{\cdot}10^6$
HPMC (λFCN) (proposed)	**96.06**	98.96	91.89	**75.82**	**93.97**	**93.36**	**96.73**	**91.68**	93.97	92.81	$17{\cdot}10^6$
HPMC (LλFCN) (proposed)	94.58	98.12	82.83	33.74	84.90	88.49	93.13	80.44	88.48	84.27	$38{\cdot}10^4$
HPMC (λCMAtt) (proposed)	83.20	93.99	76.12	48.89	76.25	91.85	88.68	78.38	72.75	75.46	$8{\cdot}10^6$
MS2CANet [18]	18.95	94.81	65.73	9.58	51.92	33.63	77.66	45.77	69.44	55.17	$2{\cdot}10^5$
Sal^{2}RN [51]	82.53	96.52	73.31	1.02	73.27	75.27	87.91	66.99	75.27	70.89	$3{\cdot}10^5$
GCFNet [45]	29.49	98.34	42.26	0.05	29.07	69.49	78.53	44.79	76.69	56.55	$11{\cdot}10^6$
PS [59] + FCN	91.20	98.32	91.20	74.78	93.05	90.48	95.54	89.84	91.59	90.70	$17{\cdot}10^6$

TABLE II

TEST-SET RESULTS FOR THE EMILIA-ROMAGNA DATASET. PER-CLASS VALUES ARE RECALLS. OVERALL ACCURACY (OA), RECALL, PRECISION, AND $F1$ SCORE ARE AVERAGED OVER THE CLASSES

Architecture	built-up	crop	tall veg.	grass	bare soil	water	OA	recall	precision	F1 score	# trainable param.
λFCN (proposed)	81.30	81.97	78.07	8.37	26.51	27.53	77.54	50.63	67.91	58.00	$17{\cdot}10^6$
LλFCN (proposed)	86.44	91.82	62.34	15.17	20.62	30.05	81.16	51.08	68.03	58.34	$38{\cdot}10^4$
λCMAtt (proposed)	70.27	93.08	38.52	32.74	32.35	34.73	75.62	50.28	58.33	54.00	$8{\cdot}10^6$
CFC-CRF [27] + λFCN	81.57	82.12	90.48	18.95	26.52	**49.40**	80.97	58.18	65.52	61.63	$17{\cdot}10^6$
CFC-CRF [27] + LλFCN	**87.03**	94.89	65.07	22.57	22.46	30.05	83.86	53.68	70.43	60.92	$38{\cdot}10^4$
CFC-CRF [27] + λCMAtt	70.86	95.07	39.37	33.41	32.51	34.89	77.11	51.02	63.11	56.42	$8{\cdot}10^6$
HPMC (λFCN) (proposed)	85.73	92.73	83.15	27.99	29.80	41.86	86.57	60.21	79.51	**68.53**	$17{\cdot}10^6$
HPMC (LλFCN) (proposed)	83.98	96.91	81.15	14.80	31.95	42.40	88.19	58.53	**80.74**	67.87	$38{\cdot}10^4$
HPMC (λCMAtt) (proposed)	71.09	94.96	**93.07**	37.73	**47.34**	36.07	**88.48**	**63.38**	66.76	65.03	$8{\cdot}10^6$
MS2CANet [18]	76.04	**97.67**	15.11	19.42	17.88	20.59	73.56	41.12	67.67	51.15	$2{\cdot}10^5$
Sal^{2}RN [51]	66.35	95.03	67.71	8.11	14.56	28.81	81.42	46.76	70.21	56.14	$3{\cdot}10^5$
GCFNet [45]	60.41	93.88	51.46	18.44	16.73	22.25	77.04	43.86	56.99	49.57	$11{\cdot}10^6$
PS [59] + FCN	75.32	90.92	60.22	**41.60**	19.74	28.53	79.38	52.72	61.46	56.76	$17{\cdot}10^6$

TABLE III

TEST-SET RESULTS FOR THE CASTILLA–LA MANCHA DATASET. PER-CLASS VALUES ARE RECALLS. OVERALL ACCURACY (OA), RECALL, PRECISION, AND $F1$ SCORE ARE AVERAGED OVER THE CLASSES

Architecture	built-up	crop	tall veg.	grass	bare soil	water	OA	recall	precision	F1 score	# trainable param.
λFCN (proposed)	61.02	99.65	97.47	92.92	71.58	71.55	97.16	82.36	91.62	86.75	$17{\cdot}10^6$
LλFCN (proposed)	42.65	98.91	95.30	85.19	55.27	50.23	94.99	71.26	88.32	78.88	$38{\cdot}10^4$
λCMAtt (proposed)	50.18	99.02	89.62	85.63	59.16	60.83	94.68	74.07	83.95	78.70	$8{\cdot}10^6$
CFC-CRF [27] + λFCN	62.25	99.66	97.58	93.11	72.88	71.50	97.26	82.83	91.99	87.17	$17{\cdot}10^6$
CFC-CRF [27] + LλFCN	44.65	98.94	95.49	85.51	57.04	50.22	95.16	71.97	88.99	79.59	$38{\cdot}10^4$
CFC-CRF [27] + λCMAtt	51.92	99.04	90.01	86.03	60.97	60.83	94.86	74.80	84.62	79.41	$8{\cdot}10^6$
HPMC (λFCN) (proposed)	**62.78**	**99.82**	**99.06**	**94.35**	**74.41**	71.59	**97.69**	**83.67**	**93.84**	**88.46**	$17{\cdot}10^6$
HPMC (LλFCN) (proposed)	60.58	99.25	97.66	85.59	58.19	50.26	96.23	75.26	92.50	82.99	$38{\cdot}10^4$
HPMC (λCMAtt) (proposed)	51.86	99.06	90.02	85.99	60.89	**99.89**	94.96	81.28	86.11	83.62	$8{\cdot}10^6$
MS2CANet [18]	18.90	97.40	71.54	52.93	16.79	0.00	87.60	42.93	57.90	49.30	$2{\cdot}10^5$
Sal^{2}RN [51]	12.02	97.16	64.60	47.71	10.04	0.31	85.90	38.64	70.84	50.00	$3{\cdot}10^5$
GCFNet [45]	16.13	97.17	64.92	54.25	22.58	34.10	86.70	48.19	72.88	58.02	$11{\cdot}10^6$
PS [59] + FCN	53.51	98.98	82.69	63.43	64.72	69.72	93.03	72.18	84.46	77.84	$17{\cdot}10^6$

The unary potentials are obtained through the neural network classifiers described in Section III-B, providing probability scores at the pixel level and feature maps at the panchromatic and hyperspectral native resolutions. The two different FCN-based and ViT-based formulations of the proposed approach were applied and tested separately. Furthermore, a lightweight variant of the FCN-based architecture (named LλFCN in the following), with fewer convolutional blocks and a pointwise channelwise spectral convolution, was also employed for experimental validation purposes. All the networks were trained on an RTX4080Ti GPU for 30 epochs, with patches of size 384×384 pixels, batch size of 10, learning rate of 0.01, and decay rate of 0.0005. The optimizer employed was the stochastic gradient descent algorithm. The overall amount of trainable parameters is shown in Tables I–III. These backbones were selected for their robustness, stability, capability of fusing cross-modal features, and encoder–decoder architecture, which allows to obtain outputs with the same size as the input [24], [54]. The proposed method involves several hyperparameters, which were experimentally set through trial and error and whose values are reported in Table IV. In particular, the weights of the pairwise potentials $\lambda_{\mathcal{SR}}, \lambda_{\mathcal{CD}}, \lambda_{\mathcal{SC}}$, were set to $2, 1, 1$, respectively. The weight γ of the cluster unary potential was set to n^2/k^p and $n^2/(\rho^2 k^h)$ in the case

TABLE IV
PARAMETERS OF THE CRF MODEL IN THE PROPOSED METHOD (THE SUPERSCRIPTS p AND h REFER TO PANCHROMATIC AND HYPERSPECTRAL, RESPECTIVELY)

Parameter	Panchromatic	Hyperspectral
$\lambda_{\mathcal{SR}}$ (pixel pairwise)	2	2
$\lambda_{\mathcal{CD}}$ (cluster pairwise)	1	1
$\lambda_{\mathcal{SC}}$ (pixel-cluster pairwise)	1	1
image patch	600×600 ($n = 600$)	100×100 ($\rho = 6$)
PRISMA Emilia Romagna dataset		
γ (cluster unary)	$n^2/k^p = 5625$	$n^2/(\rho^2 k^h) = 2500$
k (number of clusters)	64	4
h (nearest centroids)	4	4
PRISMA Lombardy dataset		
γ (cluster unary)	$n^2/k^p = 1400$	$n^2/(\rho^2 k^h) = 2500$
k (number of clusters)	256	4
h (nearest centroids)	4	4
PRISMA Castilla-La Mancha dataset		
γ (cluster unary)	$n^2/k^p = 5625$	$n^2/(\rho^2 k^h) = 2500$
k (number of clusters)	64	4
h (nearest centroids)	4	4

of the panchromatic and hyperspectral grids, respectively (details on the rationale of this setting in relation to the patch size used for CRF implementation can be found in [27]). The standard deviation σ of the Gaussian kernel in the contrast-sensitive Potts pairwise potential is the median Euclidean distance between all considered pairs of feature vectors.

After computing the multiscale tensor with the output of the network, the clustering is run on a subset of pixels to maintain a low computational complexity. The number of clusters k^h is chosen empirically as 4, with k^p varying between 256 and 64 depending on the dataset. The CRF energy minimization is implemented by subdividing the image into patches of 600×600 pixels. This allows the proposed methodology to be adapted to larger scale images, with a limited increase in computational burden. The number h of nearest centroids is set to 4 to capture long-range information while keeping a rather low computational cost.

To the authors' knowledge, this is the first method involving the cross-modal fusion and classification of panchromatic and hyperspectral images; it was not possible to perform comparisons with previous state-of-the-art methods for the same task. However, several methods have been proposed for the multimodal fusion and classification of hyperspectral and LiDAR images [18], [51] or for the fusion and classification of multispectral and panchromatic images [45]. The techniques proposed in [18], [45], and [51] were adapted for comparison purposes. In particular, the network proposed in [18], MS2CANet, involves a multiscale pyramid fusion framework based on spatial–spectral cross-modal attention to extract multiscale features and ensure interactivity of information between modes. The network in [51], Sal^{2}RN, is based on a cross-layer interaction module to integrate diversified features between various layers, on a spectrum correction module to integrate neighborhood information and reduce intraclass variance, and on a statistically based feature combination module to fuse spatial and spectral features. The method proposed in [45], GCFNet, is a fusion network for the joint classification of multispectral and panchromatic images, based on an encoder–decoder architecture developed to exploit context dependencies in the image.

Sal^{2}RN and MS2CANet were originally designed to train and test on the same scene, using disjoint subsets of isolated training and testing pixels. To ensure a fair comparison with the training approach of the proposed method, they were modified to accommodate training information in the form of patches. Specifically, dense layers were replaced with 1×1 convolutional layers to avoid flattening operations, which are suitable for isolated pixel vectors but not for image patches.

In addition, the LiDAR and hyperspectral images used for training in the original Sal^{2}RN and MS2CANet methods had the same spatial resolution, which is not the case for the considered multiresolution problem. To address this discrepancy, the multimodal contracting branches were adapted to match the available dataset sizes. In particular, the pooling layers were adjusted to have a size that is a multiple of the ratio between the spatial resolutions of the panchromatic and hyperspectral data—6 in the case of PRISMA.

GCFNet was also adapted to work with hyperspectral imagery, with a much higher number of channels than multispectral imagery, and to return pixel-wise classification maps by removing the global maxpooling layer.

The cluster fully connected CRF (CFC-CRF) method proposed in [27] was also employed as a comparison technique. It was formulated for single resolution optical imagery, so it was not applicable *per se* to the considered multiresolution problem. To adapt it to this task, it was applied to a multiscale single-resolution tensor constructed with the feature maps of the panchromatic and the hyperspectral channels resampled at a common spatial resolution (5 m) and after feature reduction. Analogously, comparisons were also performed with the combination in cascade of the neural networks within the proposed approach (λCMAtt, λFCN, and its lightweight variant LλFCN) and of this single-resolution CRF model.

Finally, a further comparison was performed with a standard FCN-based classification applied to pansharpened features. The PS method used is a single-band unsupervised model nested in a sequential band-wise adaptive scheme, where each band is pansharpened, refining the model tuned on the preceding one [59]. This method is made available by Guarino et al. [59] with a set of pretrained weights on several images of the PRISMA mission, thus being consistent with the dataset used for experiments in the present article. The result is a tensor with the same spatial resolution as the original panchromatic image and the spectral resolution of the hyperspectral image. The rationale of this comparison is to show how fused information at the native resolution performs in supervised classification tasks as compared to pansharpened data.

B. Results and Discussion

The results obtained on the considered PRISMA datasets are shown in Figs. 5–9 and in Tables I–III, in terms of recall for each class, overall accuracy, and class-averaged recall, precision, and $F1$ score.

1) Prisma Lombardy Dataset: The results obtained through the experimental validation of the method on the test set of the

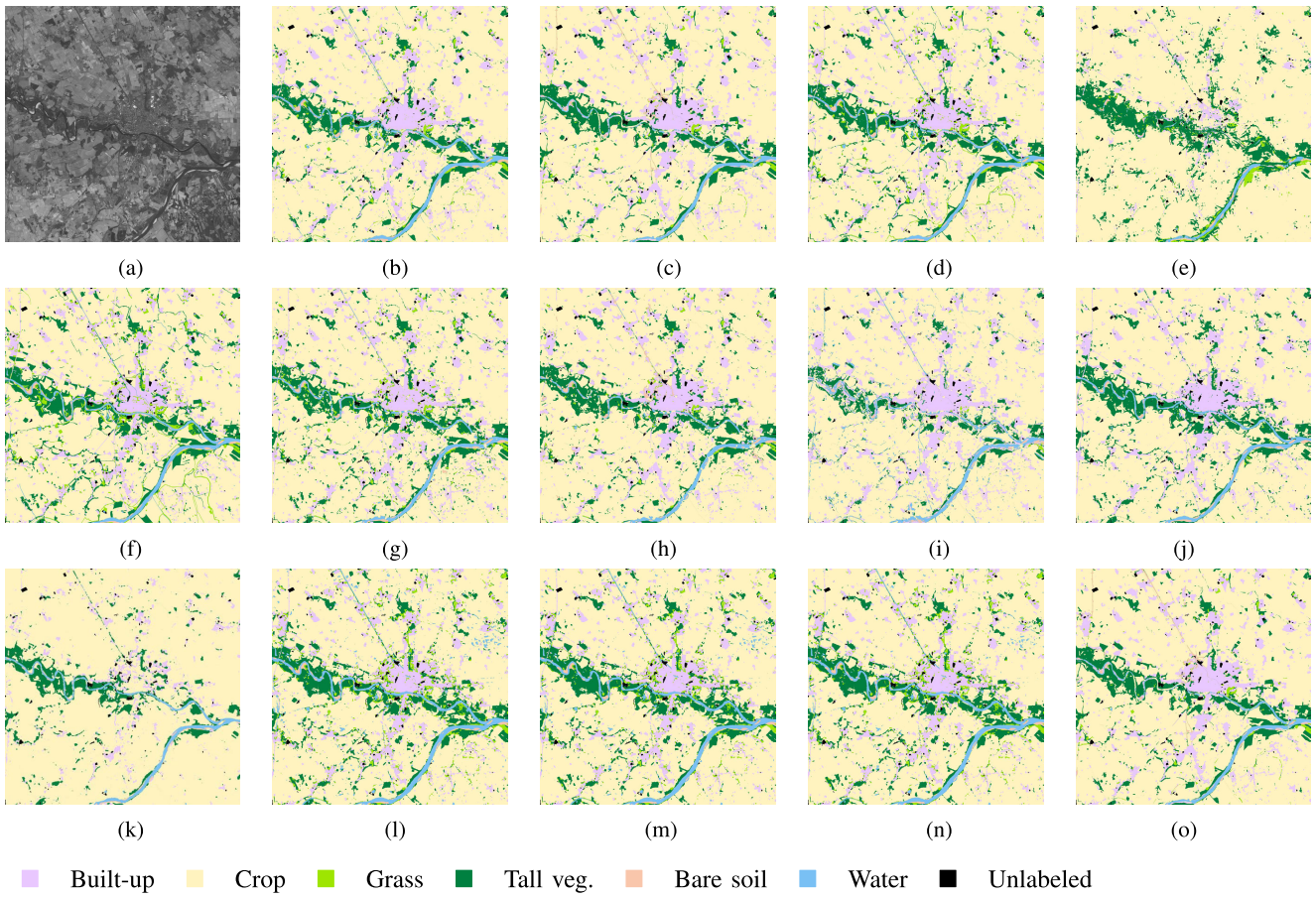


Fig. 5. Crop of test ground truth and classification maps for one test tile in the PRISMA Lombardy dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Orig. panchromatic. (b) λFCN. (c) CFC-CRF [33] +λFCN. (d) HPMC (λFCN). (e) MS2CANet [18]. (f) Ground truth. (g) LλFCN. (h) CFC-CRF [33] + LλFCN. (i) HPMC (LλFCN). (j) Sal^{2}RN [51]. (k) GCFNet [45]. (l) λCMAtt. (m) CFC-CRF [33] +λCMAtt. (n) HPMC (λCMAtt). (o) PS [59] + FCN.

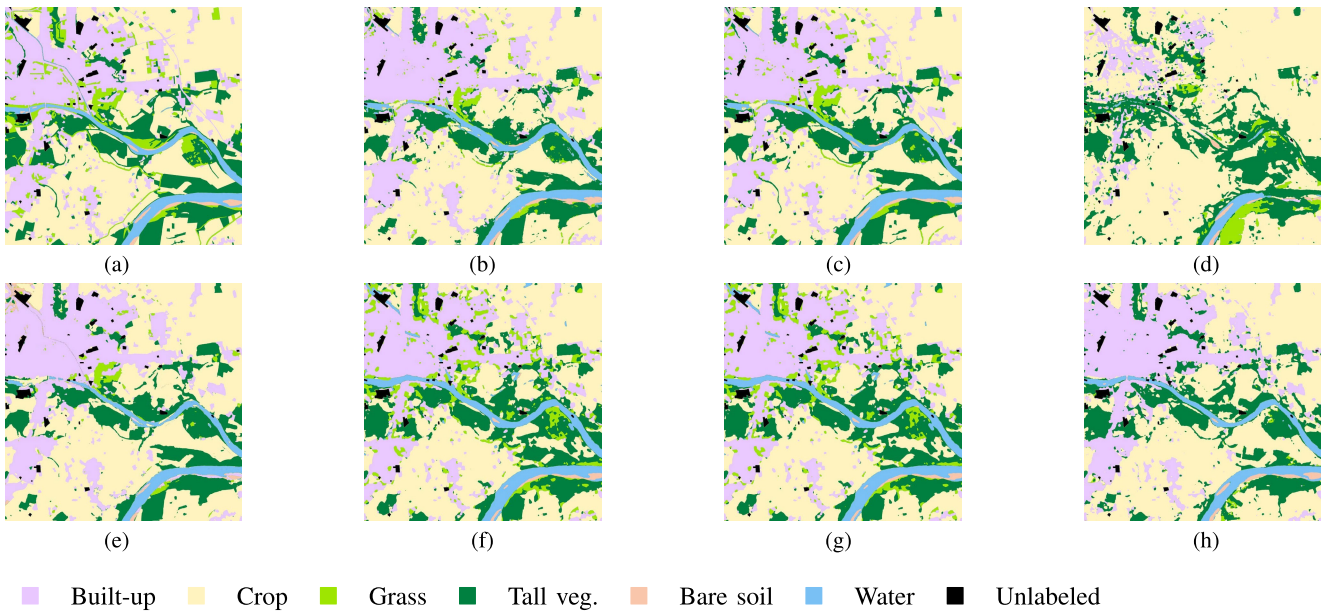


Fig. 6. Zoom-in of test ground truth and classification maps for one test tile in the PRISMA Lombardy dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Ground truth. (b) λFCN. (c) HPMC (λFCN). (d) MS2CANet [18]. (e) PS [59] + FCN. (f) λCMAtt. (g) HPMC (λCMAtt). (h) Sal^{2}RN [51].

Lombardy dataset are shown in Figs. 5 and 6 and in Table I. As can be seen from the results in Table I, the proposed neural architectures effectively distinguish the six land cover classes identified in the ground truth data. Specifically, all

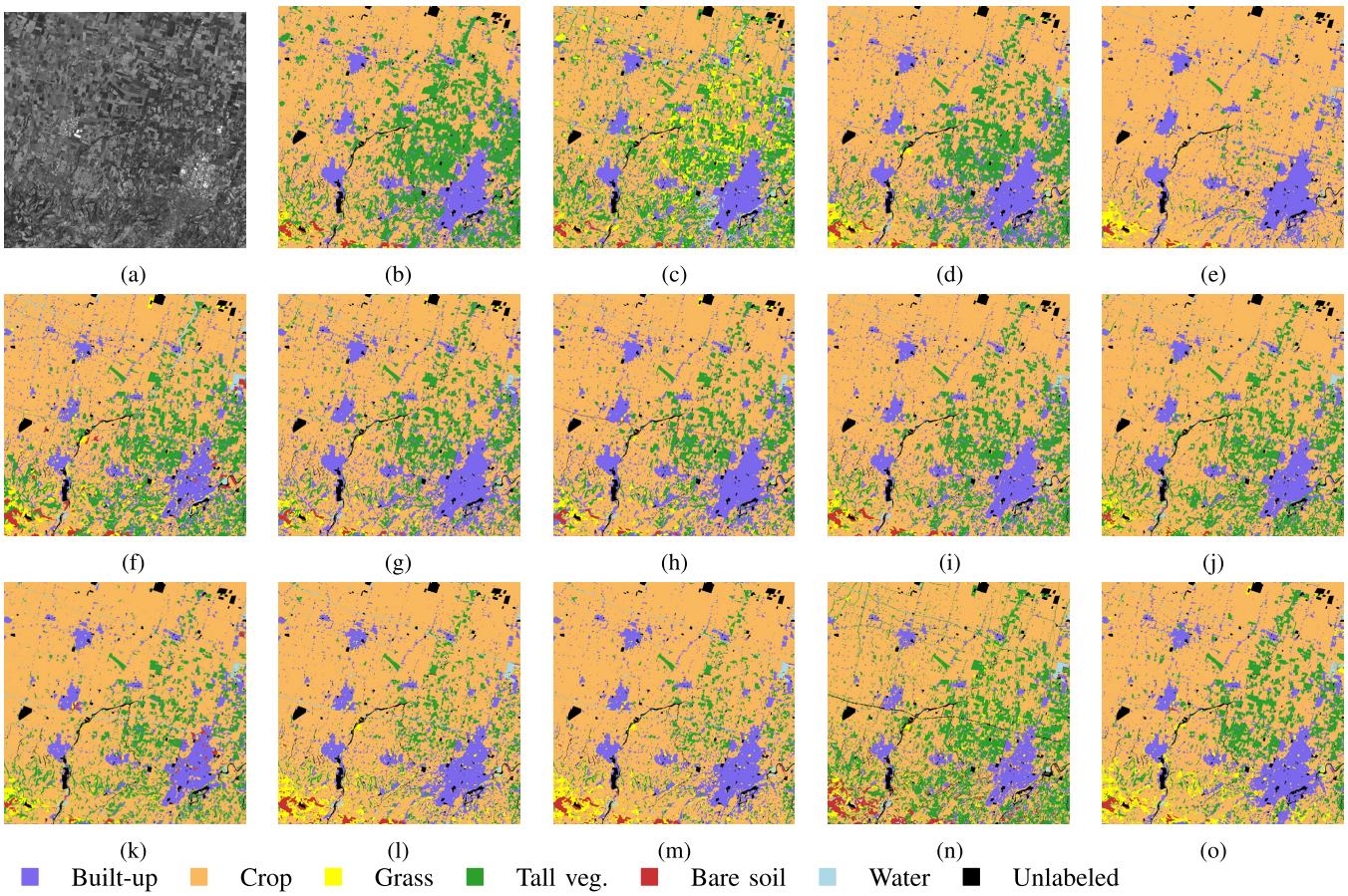


Fig. 7. Crop of test ground truth and classification maps for one test tile in the PRISMA Emilia-Romagna dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Orig. panchromatic. (b) λFCN. (c) CFC-CRF [33] +λFCN. (d) HPMC (λFCN). (e) MS2CANet [18]. (f) Ground truth. (g) LλFCN. (h) CFC-CRF [33] + LλFCN. (i) HPMC (LλFCN). (j) Sal^{2}RN [51]. (k) GCFNet [45]. (l) λCMAtt. (m) CFC-CRF [33] +λCMAtt. (n) HPMC (λCMAtt). (o) PS [59] + FCN.

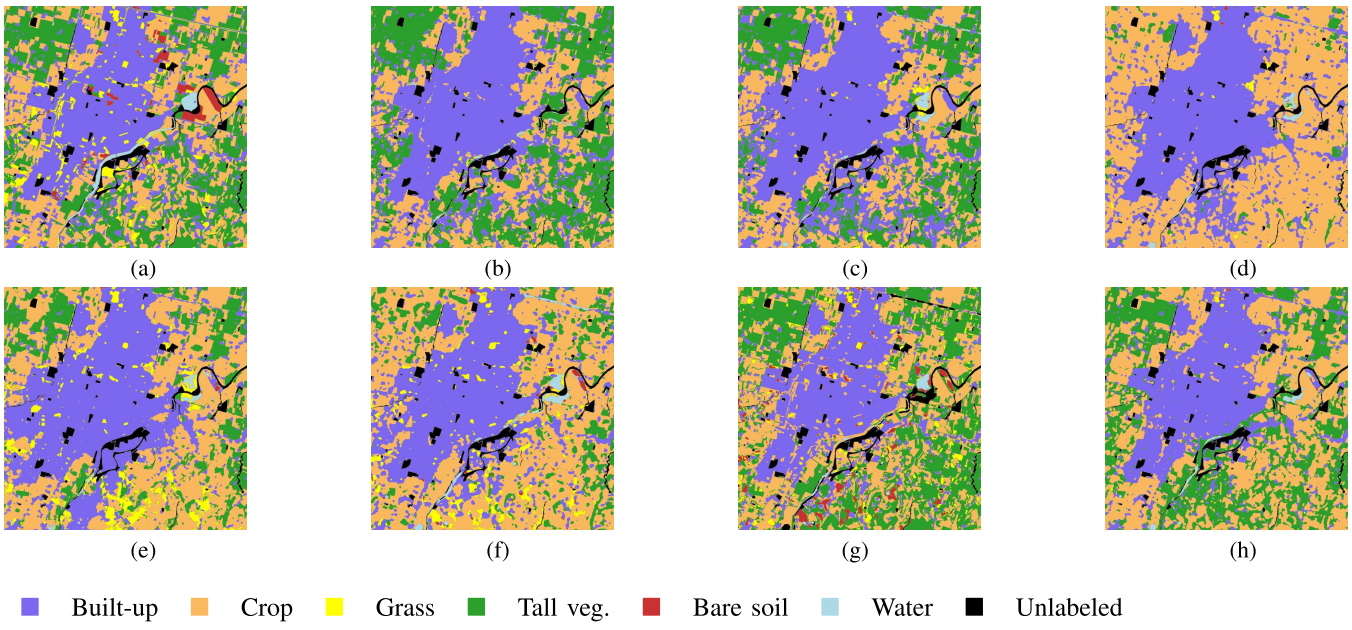


Fig. 8. Zoom-in of test ground truth and classification maps for one test tile in the PRISMA Emilia-Romagna dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Ground truth. (b) λFCN. (c) HPMC (λFCN). (d) MS2CANet [18]. (e) PS [59] + FCN. (f) λCMAtt. (g) HPMC (λCMAtt). (h) Sal^{2}RN [51].

three architectures achieve accurate results not only for the classification of majority classes such as built-up areas ("built-up") and vegetated areas ("crop" and "tall vegetation"), but also for the minority classes "bare soil" and "water."

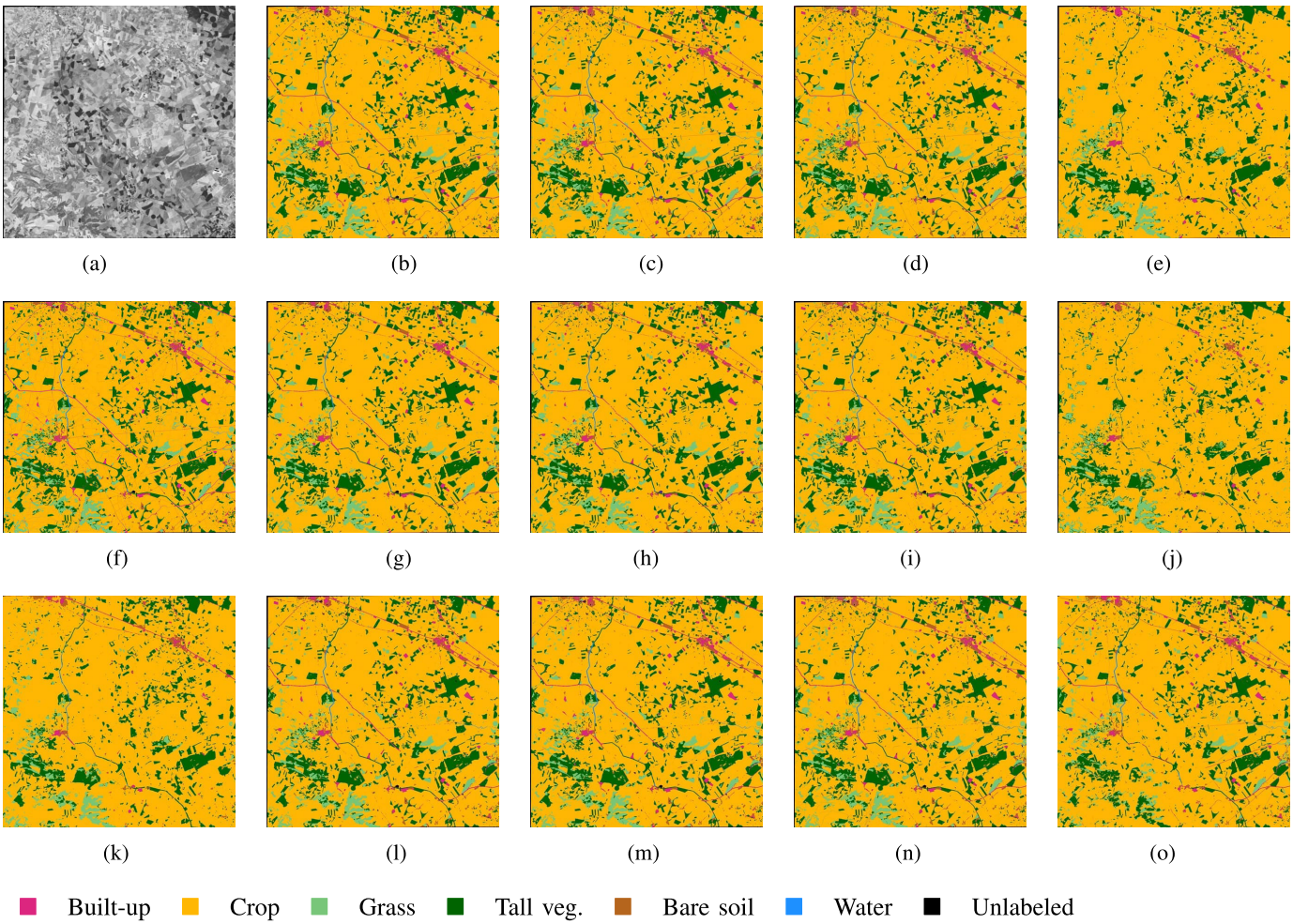


Fig. 9. Crop of test ground truth and classification maps for one test tile in the PRISMA Castilla–La Mancha dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Orig. panchromatic. (b) λFCN. (c) CFC-CRF [33] +λFCN. (d) HPMC (λFCN). (e) MS2CANet [18]. (f) Ground truth. (g) LλFCN. (h) CFC-CRF [33] + LλFCN. (i) HPMC (LλFCN). (j) Sal^{2}RN [51]. (k) GCFNet [45]. (l) λCMAtt. (m) CFC-CRF [33] +λCMAtt. (n) HPMC (λCMAtt). (o) PS [59] + FCN.

The results are slightly less accurate for the classification of low-vegetation, noncultivated areas ("grass"), primarily due to the overlap in feature space between the "tall vegetation" and "grass" classes. The "tall vegetation" class, being significantly more prevalent, is also characterized by some false positives.

As also highlighted in Table I, the inclusion of the proposed multiresolution CRF and the consequent modeling of spatial–spectral relationships, both local and long-range, improves the accuracy of results across all the considered classes and for all the proposed neural architectures, leading to enhancements in average accuracies, as compared to the application only of λFCN, LλFCN, and λCMAtt. Among these, λFCN is the architecture that achieves the best results, thanks to its higher number of parameters (which also entails a greater computational time and high sensitivity to the amount of available training samples). In general, the average values of recall, precision, and $F1$ score are around 80%, reaching up to 90% for the experiments involving λFCN.

The results obtained with the adaptation of the CRF in [27] also exhibit some improvements with respect to the proposed stand-alone multiresolution neural networks, thanks to their explicit modeling of the spatial dependencies between neighboring pixels and their spectral signatures and long-range dependencies through the preliminary clustering partition. Yet, the multiresolution CRF formulation developed in the present article and integrated in the proposed method allows for more accurate results in terms of recall, and similar results in terms of overall accuracy, $F1$ score, and precision, while eliminating the need for the upsampling step.

As shown in Fig. 5, and in the zoom-ins in Fig. 6, the results obtained by the proposed method closely reflect the ground truth, with no significant oversmoothing or evident false positives. Notably, the proposed method demonstrates its ability to generate classification maps where the spatial boundaries between classes are well-defined and effectively identifies minority classes such as "bare soil," "water," and "grass."

Regarding comparisons with other state-of-the-art techniques, the results obtained by the FCN applied to the pansharpened features resulting from the approach in [59] (PS + FCN) achieve the second-best performance after all experiments involving λFCN. As seen in the last column

of Table I, these two architectures are characterized by a higher number of parameters, which explains their superior performance, but demands more resources in terms of training data and computational capabilities. Sal^{2}RN [51] achieves lower accuracy on this dataset while maintaining a high overall accuracy. The results for MS2CANet indicate lower recall for the classes "built-up" and "tall vegetation," resulting in lower overall accuracy values. GCFNet [45] obtains results similar to those of MS2CANet, with a slightly higher OA, higher precision, but lower recall. As shown in Table I, higher performance figures are obtained by the proposed approach than by the considered benchmark techniques.

2) PRISMA Emilia-Romagna Dataset: For the Emilia-Romagna dataset, the results are shown in Figs. 7 and 8 and Table II. As can be seen from the results in Table II, the proposed method, with both neural architectures, effectively distinguishes the three primary land cover classes: built-up areas ("built-up") and vegetated areas ("crop" and "tall vegetation"). The results are generally less accurate for the minority classes, "bare soil," "grass," and "water." This behavior can be mainly attributed to dataset-related factors. In particular, the ground truth for the Emilia-Romagna dataset is derived from a land-use archive characterized by a coarser spatial resolution than the panchromatic imagery, which introduces spatial inaccuracies in the class boundaries and especially affects small or fragmented regions. Moreover, the temporal mismatch between the acquisition of the reference ground truth data (2020) and the PRISMA imagery (2022) leads to inconsistencies in classes subject to temporal variations, such as low vegetation and water bodies. Finally, the limited number of training samples available for minority classes further contributes to misclassifications and confusion between spectrally similar categories. Concerning the stand-alone neural networks, LλFCN obtains slightly more accurate results.

Both in the case of the proposed approach and in the case of the application of the adapted CFC-CRF after the proposed networks, the use of CRFs resulted in improvements in the accuracy of all considered classes and in the average accuracy. In particular, higher overall classification accuracies can be noted with the proposed HPMC method, leading to an overall accuracy of 88%, precision of 80%, and $F1$ score of about 70%, despite the aforementioned issues remarked in the dataset. In general, the results of the proposed multiresolution approach, in its three variants associated with distinct neural architectures, are more accurate in terms of all the averaged accuracy metrics.

Fig. 7 displays the classification maps obtained using the proposed and the comparison techniques (Fig. 8 presents a zoomed-in detail of such maps). The results obtained by the method closely reflect the ground truth, particularly for the majority classes, with no significant oversmoothing or evident false positives (except for the class "grass" with λFCN + [27]). Notably, the proposed method demonstrates its ability to generate classification maps where the spatial boundaries between classes are well-defined. λCMAtt exhibits some underclassification of the class "tall vegetation," both for the stand-alone version and when applied together with CFC-CRF. On the contrary, the discrimination of this class is recovered by the developed approach. In the case of the Emilia-Romagna dataset, for which the results are overall less accurate, the improvements provided by the proposed HPMC technique are particularly evident.

The results obtained by Sal^{2}RN [51], GCFNet [45], and PS [59] + FCN are comparable to the ones obtained by the three proposed neural architectures in their stand-alone application. In particular, for this dataset, the combination PS + FCN obtains the most accurate results for the class "grass," one of the minority classes in the dataset. The results for MS2CANet exhibit a lower accuracy for the class "tall veg.," which influences the averaged accuracy metrics accordingly.

3) PRISMA Castilla–La Mancha Dataset: For the Castilla–La Mancha dataset, the quantitative and qualitative results are reported in Figs. 9 and 10 and Table III. As shown in Table III, all the variants of the proposed approach achieve very high accuracies for the majority classes, in particular for "crop" and "tall vegetation," for which recalls above 97% and 90%, respectively, are consistently obtained. The class "grass" is also accurately identified by the proposed method, with recalls exceeding 85% for all architectures. Conversely, lower accuracies are observed for the classes "built-up," "bare soil," and "water," which represent more challenging categories due to their higher intraclass variability (especially for "built-up" and "bare soil") and a lower number of training samples, given the prevalence of vegetated areas for this dataset.

Compared to the Emilia-Romagna dataset, the overall performance on Castilla–La Mancha is higher (similar to the Lombardy dataset), as reflected by overall accuracies close to or above 97% for the best-performing configurations. Among the stand-alone networks, λFCN provides the most accurate results, achieving an OA of 97.16% and an average $F1$ score of 86.75%, while also demonstrating strong robustness across all classes. LλFCN, despite its significantly reduced number of trainable parameters, exhibits only slightly lower performance, particularly for the minority classes, highlighting the tradeoff between model complexity and classification accuracy.

The proposed multiresolution HPMC approach leads to systematic, slight gains over both the stand-alone networks and the single-resolution CFC-CRF model. This improvement is evident across nearly all classes and architectures, with the most notable enhancements observed for "built-up," "bare soil," and "grass." For instance, in its version with λFCN, the proposed approach reaches an OA of 97.69%, with corresponding gains, as compared to λFCN itself, in recall, precision, and $F1$ score, for which average values of 83.67%, 93.84%, and 88.46%, respectively, are obtained. The confusion matrices in Fig. 11 further confirm that the proposed multiresolution strategy tends to reduce class confusion, particularly between spectrally similar categories such as "grass" and "crop," and in general to improve the classification of all the classes which are not as majoritarian as "crop."

Figs. 9 and 10 qualitatively illustrate the classification maps obtained by the proposed and benchmark techniques. Like in the case of the other datasets, the proposed HPMC

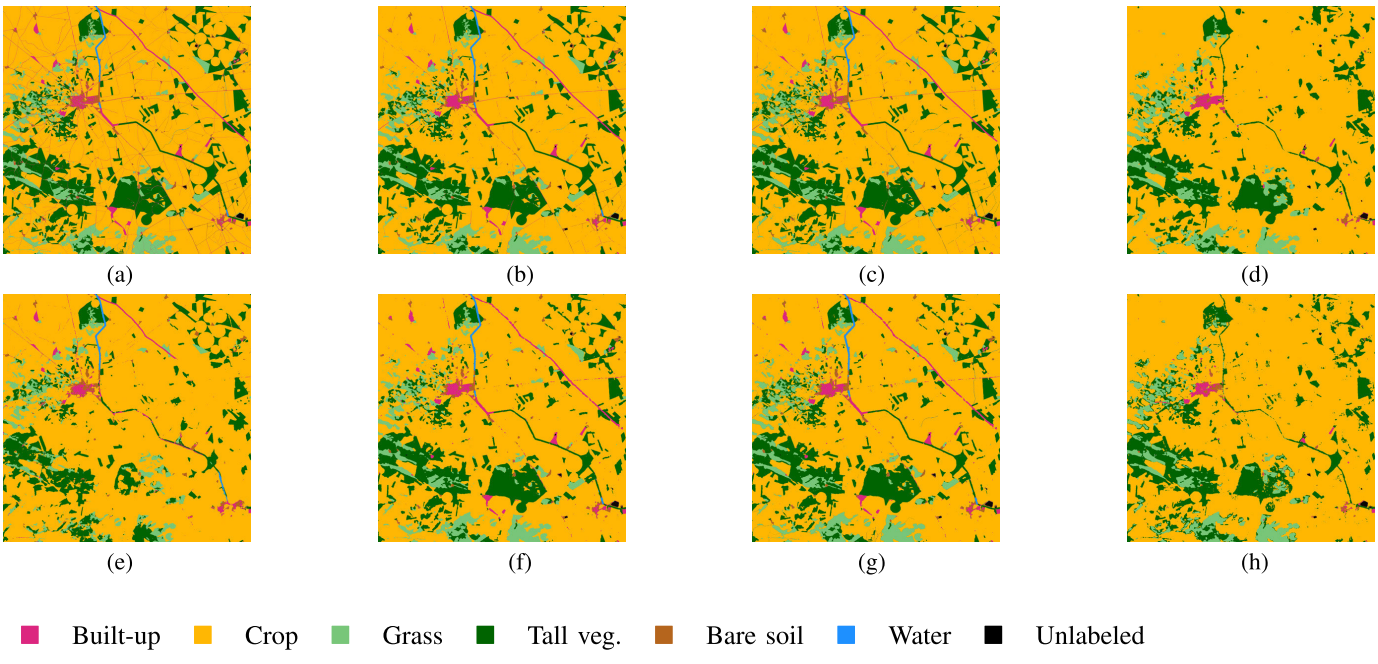


Fig. 10. Zoom-in of test ground truth and classification maps for one test tile in the PRISMA Castilla–La Mancha dataset. Original PRISMA product and product processed under a license of ASI © ASI (2022). (a) Ground truth. (b) λFCN. (c) HPMC (λFCN). (d) MS2CANet [18]. (e) PS [59] + FCN. (f) λCMAtt. (g) HPMC (λCMAtt). (h) Sal^{2}RN [51].

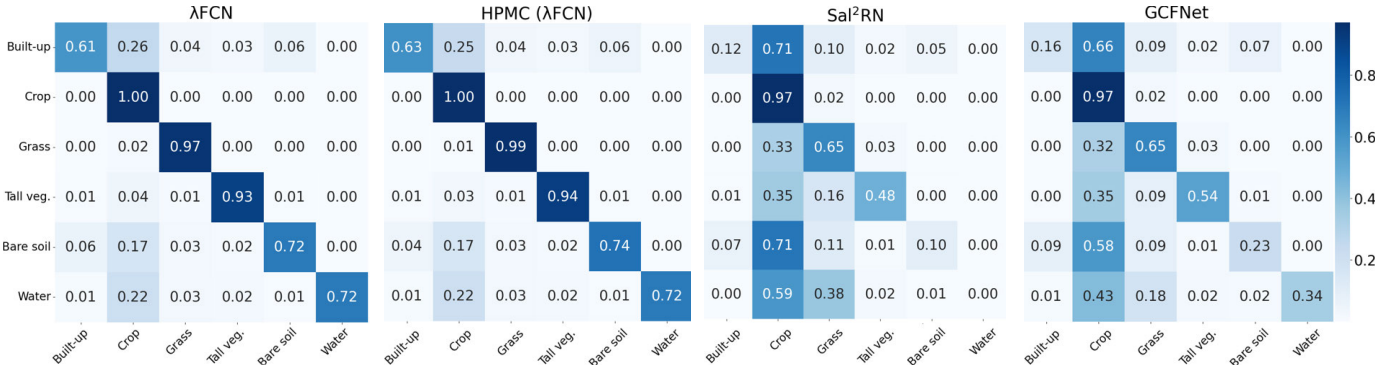


Fig. 11. Confusion matrices for the proposed method in its HPMC (λFCN) version, for the standalone λFCN component (i.e., without the subsequent CRF-based stage), and for Sal^{2}RN and GCFNet, evaluated on the PRISMA Castilla–La Mancha dataset. The rows and columns correspond to the true and predicted values, respectively.

method produces spatially coherent maps that closely match the ground truth, with well-defined class boundaries and limited over-smoothing effects. Compared to the adaptation of the single-resolution CFC-CRF model, the multiresolution CRF formulation in HPMC better preserves fine spatial details while maintaining global consistency, particularly in heterogeneous areas and along spatial class transitions. The zoomed-in views in Fig. 10 highlight the improved delineation of small structures and fragmented regions, especially for "built-up" areas and narrow water bodies.

Regarding the other comparison techniques, MS2CANet, Sal^{2}RN, and GCFNet obtain lower performance on this dataset, especially for minority classes such as "built-up," "bare soil," and "water" (see also the confusion matrices in Fig. 11), which substantially affects their averaged accuracy metrics. The combination PS + FCN achieves competitive results among the comparison methods, particularly for "built-up" and "bare soil," although the proposed approach obtains more accurate results in all its versions. Overall, these results confirm that the joint use of the proposed neural architectures and multiresolution CRF model for hyperspectral and panchromatic imagery yields accurate and spatially consistent classification maps for the Castilla–La Mancha dataset as well.

4) Statistical Significance of the Differences Among the Results: The McNemar's test was employed to validate whether the difference in accuracy between the proposed HPMC method and each technique considered for comparison was statistically significant. The test computes the asymptotically normal statistics: $Z = (f_{12} - f_{21})/((f_{12} + f_{21})^{1/2})$, with f_{12} being the number of test samples misclassified by the proposed method and not by the comparison method, and f_{21} being its opposite [60], [61]. Using the common 5% level of significance, the difference in results is statistically significant if $|Z| > 1.96$: in particular, a negative value of Z indicates that

TABLE V

Z STATISTICS OF THE MCNEMAR TEST TO VALIDATE WHETHER THE DIFFERENCES IN ACCURACY ARE STATISTICALLY SIGNIFICANT: FOR EACH DATASET, THE VALUE OF *Z* CORRESPONDS TO THE RESULT OF THE COMPARISON BETWEEN THE PROPOSED METHOD AND EACH BENCHMARK TECHNIQUE

	λCMAtt	**HPMC (λCMAtt)**	**λFCN**	**HPMC (λFCN)**	**Sal^{2}RN**	**MS2CANet**	**PS + FCN**	**GCFNet**
PRISMA Lombardy [HPMC (λFCN)]	-82.10	-69.35	-90.69	—	-50.15	-68.07	-31.77	-55.25
PRISMA Lombardy [HPMC (λCMAtt)]	-46.69	—	16.98	69.35	-11.24	-23.33	52.02	-71.68
PRISMA Emilia Romagna [HPMC (λFCN)]	-114.18	13.05	-143.00	—	-58.92	-116.15	-92.93	-102.09
PRISMA Emilia Romagna [HPMC (λCMAtt)]	-158.46	—	-105.18	-13.05	-68.11	-119.08	-95.68	-108.18
PRISMA Castilla-La Mancha [HPMC (λFCN)]	-67.37	-25.98	-93.31	—	-102.02	-88.78	-1.58	-100.52
PRISMA Castilla-La Mancha [HPMC (λCMAtt)]	-24.03	—	32.20	25.98	-94.01	-72.08	-42.04	-135.03

the proposed method is more accurate than the other used for comparison [61].

The outcome of McNemar's test is reported in Table V. For each dataset, the test compares the proposed HPMC approach, in either the λFCN or λCMAtt versions, with the corresponding stand-alone neural architecture and with the considered benchmark techniques (the outcome of the test in the case of LλFCN is analogous and is omitted for brevity). As expected, the absolute values of *Z* obtained when comparing different variants of the proposed method among one another (e.g., ViT-based versus FCN-based architectures) are generally lower than those observed in the comparisons with external baselines. This behavior is expected and is attributed to the fact that these versions are all consistently developed to address the problem of multiresolution hyperspectral-panchromatic classification, hence they share a similar spatial–spectral fusion and regularization strategy, leading to more subtle performance differences.

Conversely, large absolute values of *Z* are consistently observed when comparing the proposed approach with the benchmark techniques across all three datasets, with $|Z|$ frequently exceeding 50, confirming that the accuracy gains achieved by the developed method are statistically significant.

Although several of the benchmark methods already provide competitive results, the outcome of McNemar's test confirms that the improvements introduced by the proposed HPMC approach are not due to random variations but reflect a statistically significant enhancement in classification accuracy. This holds for both ViT-based and FCN-based configurations and across datasets with different characteristics and levels of class imbalance. Overall, the statistical analysis corroborates the effectiveness of the proposed approach.

V. CONCLUSION

This article proposes a novel multiresolution supervised method for the joint fusion and classification of panchromatic and hyperspectral images. The proposed method combines a cross-modal fusion neural architecture, based on FCNs or ViTs, with a multiresolution CRF to model the spatial and spectral information provided by the multiresolution and hyperspectral input data and to generate a classification map at the finer of the two observed spatial resolutions. The neural networks in the various versions of the proposed method characterize multiresolution information through dedicated encoder–decoder architectures, and leverage the spectral information in the hyperspectral data through suitable 3-D operators. The CRF within the developed approach models the multiresolution relationships between the hyperspectral and panchromatic images as well as short-range and long-range spatial interactions in each image, by combining a multiresolution graph with fully connected subgraphs associated with clustering partitions.

The method was applied to PRISMA images, in case studies of land cover mapping in mainly urban zones over Northern and Central Italy, and in a vegetated area in Spain. The results of the experiments demonstrate the effectiveness of the proposed method in jointly classifying panchromatic-hyperspectral imagery, with particular emphasis on its ability to model both spatial and spectral relationships. The inclusion of the multiresolution fully connected CRFs further improves classification accuracy, as compared to using only fully convolutional or attention-based neural components, by refining spatial boundaries and incorporating long-range dependencies, particularly for minority classes. The comparison with state-of-the-art techniques for cross-modal fusion and semantic segmentation of hyperspectral data, originally developed either for the combination with LiDAR data or for multispectral imagery and adapted to the case of panchromatic data, and with a PS technique, confirmed the potential of the proposed approach.

The proposed method is developed for application to single-mission hyperspectral-panchromatic images that are assumed to be registered or exhibit only minimal misalignment. On one hand, this working hypothesis is important because the multiresolution formulation of the method assumes that corresponding pixels represent the same ground locations in the hyperspectral and panchromatic images. On the other hand, various registration techniques can be employed to enforce this condition, ranging from classical feature-based and intensity-based approaches to recent DL-based techniques [62], [63], [64].

While the proposed method achieves accurate overall results, often classifying minority classes with higher accuracy than the considered benchmark techniques, the discrimination of such classes still remains a challenge in datasets with limited training samples or outdated ground truth data. These findings underline the importance of aligning data sources and employing models that can handle temporal inconsistencies and class imbalances. Future work will focus on improving the robustness of the proposed framework against such issues, integrating more advanced contextual models and semi-supervised techniques to address sample scarcity [65].

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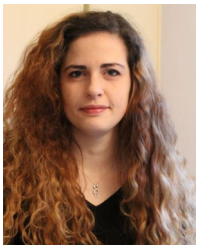
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